

A SPECIAL SWITCH For The Specially Aabled

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Specially Aabled persons, without hands or feet, often find it difficult to switch on an appliance. This switch can come to their rescue. It can be operated with a torch light or a laser light by those who are unable to use their feet, and with a footswitch by those who are unable to use their hands, to operate home appliances, fixtures, toys, learning devices, and other gadgets. The author's circuit wired on a breadboard is shown in Fig. 1.

Circuit diagram of the switch for the specially abled is shown in Fig. 2.

The circuit comprises a bridge rectifier (BR1), 12V voltage regulator 7812 (IC1), dual flip-flop 4027 IC (IC2), 12V one-changeover relay (RL1), a foot switch (S1), and a few other components. Capacitors connected across the supply terminals are used to minimise any noise signals.

A mains 230V AC primary to 15V, 500mA secondary transformer (not shown in Fig. 2) may be used to step down the AC mains voltage to 15V AC. This voltage, connected across CON1 in the circuit, is rectified by bridge rectifier BR1 and

filtered by capacitors C1 and C2 to remove any ripples in the voltage. The rectified and filtered voltage is then regulated by IC 7812 (IC1) to provide 12V DC to operate the circuit.

The LDR's resistance varies with the intensity of light falling on it. When the light falling on LDR increases, the LDR resistance decreases, and when the light falling on LDR decreases, its resistance increases. In darkness, when there is no light, the resistance of LDR is in the range of mega ohms, but in presence of bright light it decreases to a few hundred ohms.

IC 4027 has flip-flops IC2A and IC2B that are used for data storage. IC2C is its power supply terminal. The square wave generated by flip-flop IC2B is coupled to flip-flop IC2A, whose output is given to relay driver transistor T1 to drive relay RL1.

When a foot-disabled person focuses torch or laser light on the LDR, it triggers flip-flop IC2 and its

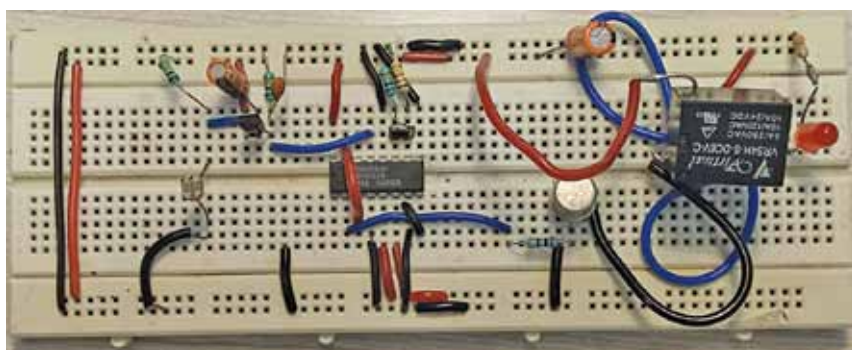


Fig. 1: Author's circuit wired on a breadboard

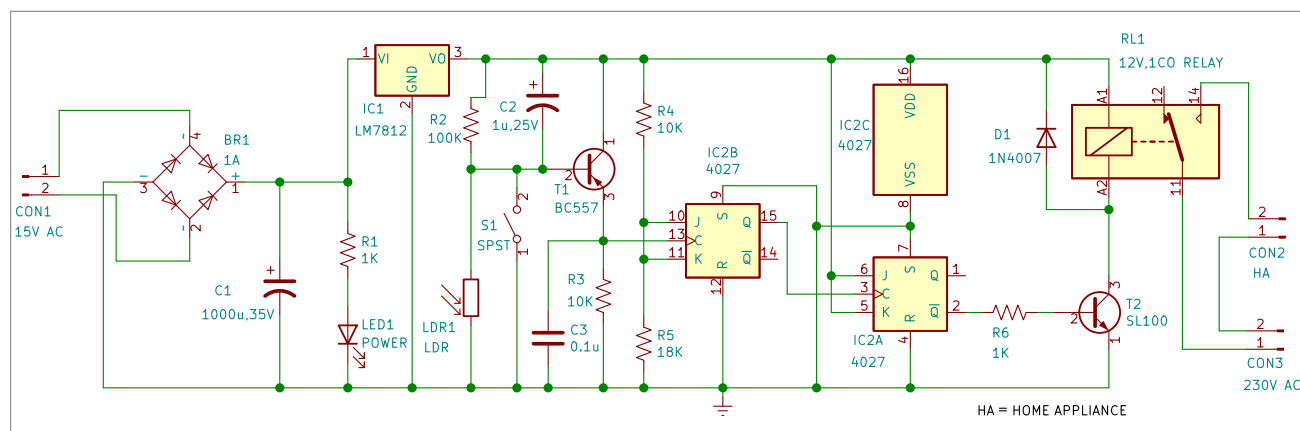


Fig. 2: Circuit diagram of the switch